INCH-POUND

A-A-59279A(AR)
18 August 2026
SUPERSEDING
A-A-59279
13 July 1998

# COMMERCIAL ITEM DESCRIPTION

## SODIUM SULFATE, ANHYDROUS, TECHNICAL (FOR AMMUNITION USE)

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for anhydrous sodium sulfate used for the manufacturing of propellants used in ammunition.

2. SALIENT CHARACTERISTICS.

2.1 Materials. The anhydrous sodium sulfate shall be in the form of a white powder when visually examined and shall be free from lumps, caking, and visual impurities. The anhydrous sodium sulfate shall be provided in the quantity specified in the contract or order (see 6.2).

2.2 Loss on ignition. The moisture content ($H_2O$) of the anhydrous sodium sulfate shall not exceed 0.5 percent when tested in accordance with American Chemical Society (ACS) publication, "Reagent Chemicals" (see 6.6.a). Loss on ignition would include the moisture content ($H_2O$) and other volatiles or trace impurities.

2.3 Assay. The anhydrous sodium sulfate shall not be less than 98.0 percent pure when tested in accordance with American Chemical Society (ACS) Specifications, "Reagent Chemicals" (see 6.6.b).

2.4 Neutrality. The acidity shall be not more than 0.005 percent calculated as sulfuric acid, or the alkalinity shall be not more than 0.5 percent calculated as sodium carbonate when tested in accordance with paragraph 4.2.

2.5 Chlorides. The chloride content shall not exceed 0.7 percent by weight when tested in accordance with paragraph 4.3.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.6 Insoluble matter. The insoluble matter shall not exceed 0.1 percent by weight when tested in accordance with American Chemical Society (ACS) Specifications, "Reagent Chemicals" (see 6.6.c).

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Neutrality test. Dissolve a 10.00 gram sample in 200 mL of carbon dioxide free water and add 10 drops of 0.1% methyl red indicator. If the solution is acidic (not purely yellow, but orange or pink), titrate with 0.01 N sodium hydroxide to a stable, pure yellow that does not change significantly with additional titrant. Calculate the acidity as follows:

$$\text{Percent sodium sulfuric acid} = \frac{4.9 \times N \times V}{W}$$

Where: N = normality of sodium hydroxide solution
V = volume of sodium hydroxide solution
W = weight of sample, grams

If the solution is basic (pure yellow), titrate with 0.1 N hydrochloric acid to a stable, pure pink that does not change significantly with additional titrant. Calculate alkalinity as follows:

$$\text{Percent sodium carbonate} = \frac{5.3 \times N \times V}{W}$$

Where: N = normality of hydrochloric acid
V = volume of hydrochloric acid
W = weight of sample, grams

4.3 Chlorides test. Chlorides shall be determined turbidimetrically after precipitation with silver nitrate.

4.3.1 Reagents and stock solutions. All materials used shall be reagent grade or as specified below.

a. Water, deionized or distilled, conforming to ASTM D1193 Type I or II
b. Nitric Acid (HNO3), concentrated, 67-70 %wt
c. Silver nitrate, 3 wt%, acidified to pH 1 with nitric acid: dissolve 0.3 g silver nitrate in 10 mL water and add 0.1 mL of nitric acid.



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d. Chloride stock, 63 mg/L: transfer 104 ± 2 mg of sodium chloride to a 1 L volumetric flask, dilute to volume with water and mix well.

4.3.2 Preparation of sample. Prepare two replicates as follows. Dissolve 180 ± 2 mg of the sodium sulfate sample in water, and dilute the solution to 50 mL in a 50-mL volumetric flask. Transfer a 1.00 mL aliquot of this solution to a 50 mL Nessler tube. Add 40-45 mL of water. Prepare the second replicate in the same manner.

4.3.3 Preparation of standard solutions. Add 40-45 mL water to each of four (4) clean 50-mL Nessler tubes. Add 0.30, 0.40, and 0.50-mL aliquots of the chloride stock to the 1st, 2nd, and 3rd tubes, respectively. The standards correspond to, respectively, 0.53, 0.70, and 0.87 weight percent (wt%) chloride. The fourth tube serves as the reagent blank.

4.3.4 Analysis. To each of the six (6) Nessler tubes (two samples, three standards, and blank) add 1.0 mL of nitric acid and 1.0 mL of 3 wt% silver nitrate. Make up to volume with water, mix well and leave in the dark for five minutes. Examine the relative opacities of the solutions by looking lengthwise through the Nessler tubes against a dark matte background; allow for adequate lighting but avoid bright lights and reflections. For each of the two replicates compare its opacity with those of the standards and identify the closest standard that is more opaque than the sample. This represents the upper limit for chloride content. The sample opacity shall be not greater than that of the 0.70 wt% Cl standard.
Note: The two sample solutions may have slightly different opacities, but both must have the same upper limit. The blank shall be less opaque (or altogether clear) than all other solutions. Otherwise, the analysis shall be repeated ensuring all glassware has been thoroughly cleaned.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 The Code of Federal Regulations (CFR) is available by mail order from the Superintendent of Documents, ATTN: New Order, P.O. Box 371954, Pittsburgh, PA 15250-7954.

6.1.2 ACS Specifications publication is available from American Chemical Society, Publications Support Services, 1155 Sixteenth Street, NW, Washington DC 20036, or online at https://pubs.acs.org/doi/10.1021/acsreagents.4368.20251001.

6.1.3 ASTM standards are available from https://www.astm.org , or ASTM International, 100 Barr Harbor Dr., W. Conshohocken, PA, 19428-2959.

6.1.4 Federal standards are available online at https://www.quicksearch.dla.mil .

6.1.5 Federal acquisition regulation (FAR) standards is available online at https://www.acquisition.gov/browse/index/far .



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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this CID.
b. Quantity required (see 2.1).
c. Packaging and marking requirements (see 5).

6.3 Material Safety Data Sheets. Products meeting the requirements of this CID may be subject to the requirements of 29 CFR 1910.1200 and FED-STD-313 for Material Safety Data Sheets (MSDS).

6.4 Transportation. Anhydrous sodium sulfate is subject to the requirements of 49 CFR 100-185 regarding the transportation of hazardous materials.

6.5 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the commercial item description. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM, Armaments Center, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.6 Test recommendations.

a. For loss on ignition (2.2), it is recommended to prepare crucibles by burning them out at 800ºC for 30-60 mins prior to cooling and obtaining tare weight.

b. For assay (2.3), fresh resin is recommended for every half-dozen samples and should be replaced as soon as assays begin to yield significantly lower or failing results. Care should be taken during NaOH titrant standardization: either commercial 0.1N NaOH should be acquired, or if standardization is done in-lab against KHP with phenolphthalein indicator, the titration endpoint should be taken as a very light pink which is typically 0.05-0.10 mL past the first hint of color; it may also be verified potentiometrically with a pH electrode.

c. For insoluble matter (2.6), it is recommended to prepare crucibles by washing them with hot water prior to drying and taring.

6.7 Changes incorporated. This specification was updated from A-A-59279 to A-A-59279A as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results. Reference documents have been verified, updated, and hyperlinked to sources. Details of these changes are documented in the ECPs and ERR listed below.

| Specification ID | ECP / ERR |
|---|---|
| A-A-59279 | |
| A-A-59279A | Y26Q2001 |

6.8 Keywords.

Chemical
Deterrent
Propellant
Munitions Chemical Specification Modernization (MCSM)
Small caliber



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CONCLUDING MATERIAL

Custodian:
Army-AR

Review activities:
Army-AV, MD1
DLA-GS

Civil agency:
OT - FSS

Preparing activity
Army -AR
Project 6810-2026-001



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